

SHETH L.U.J. AND SIR M.V. DEGREE COLLEGE OF SCIENCE

SUBJECT NAME: Power BI And Tableau Practical PRACTICAL - 9

AIM: Cleaning and Structuring Data for Analysis

- a) Identify and handle missing or inconsistent data
- b) Convert wide data into tall format
- c) Perform union of multiple files
- d) Execute cross-database joins
- e) Restructure data for better visualisation
- f) Handle different levels of detail in datasets
- g) Apply advanced data cleaning techniques

Dataset: Custom Designed Students Dataset [Sem1_Scores.csv, Sem2_Scores.csv, Department_Master.csv and Student_Activities.csv]

Steps:-

A: Identify and Handle Missing or Inconsistent Data

Step 1: Load the Dataset

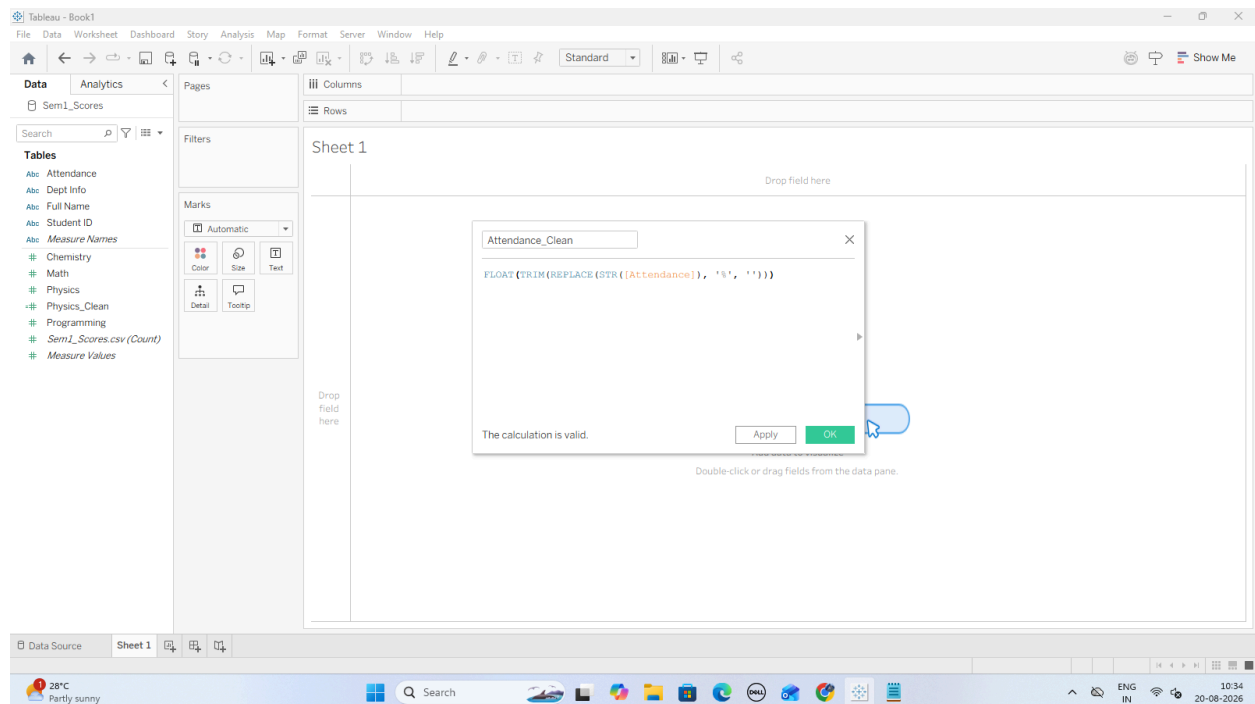
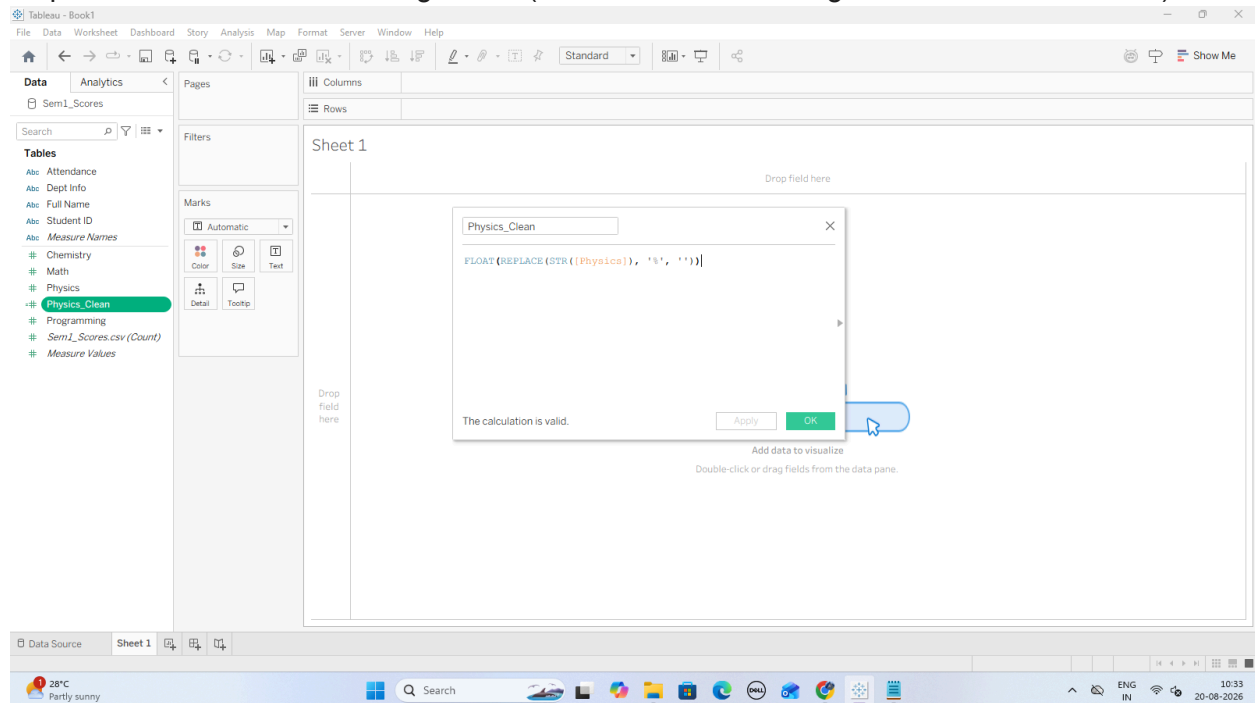
The screenshot shows the Tableau Desktop interface with the 'Sem1_Scores' dataset loaded. The sidebar on the left lists connections and files. The main view displays the dataset fields and a preview of the data.

Name	Sem1_Scores.csv
Student ID	85
Full Name	Aarav Sharma
Dept Info	CSE-01
Math	0.900000
Physics	0.750000
Chemistry	0.820000
Programming	0.900000
Attendance	88

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Step 2: Clean Inconsistent String Fields (Convert Text Percentages to Numeric Measures)



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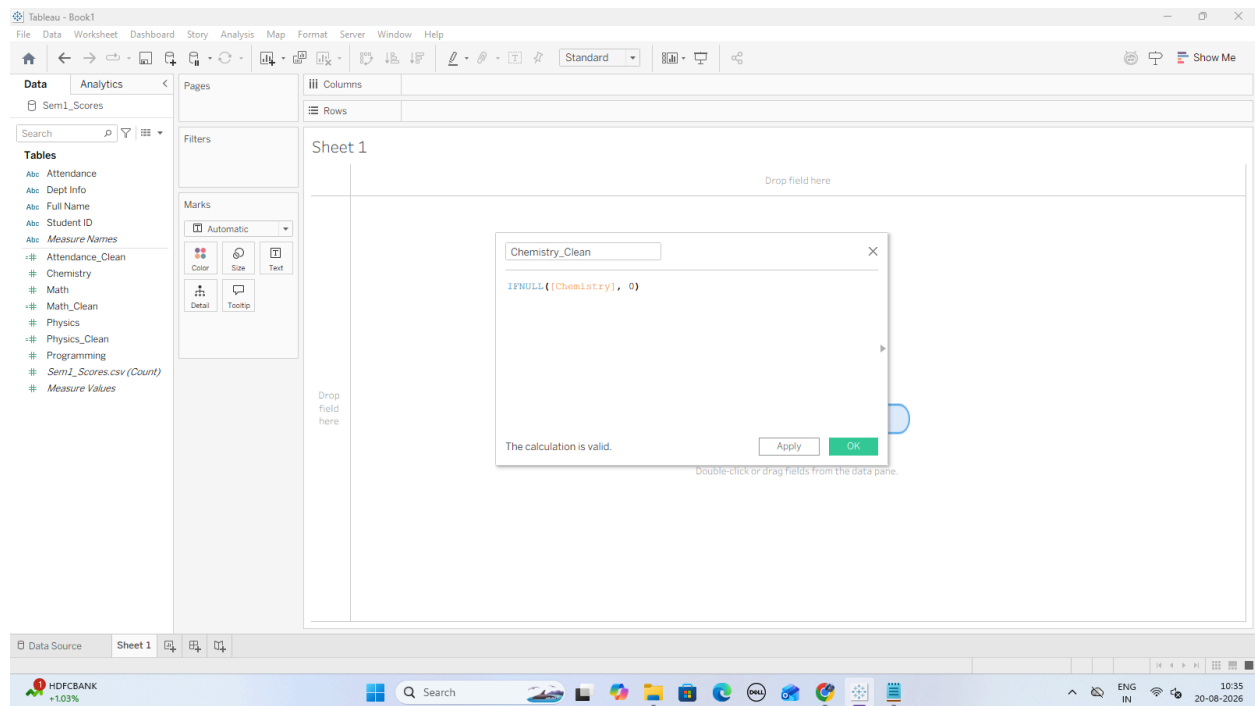
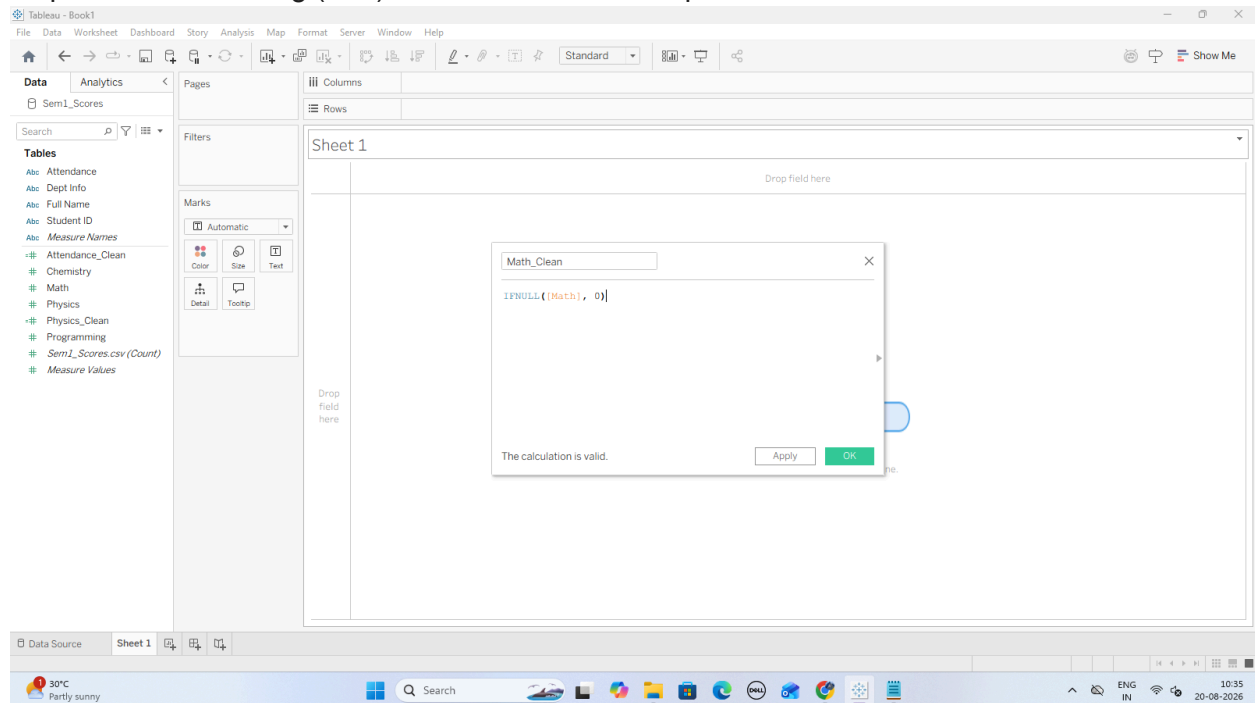
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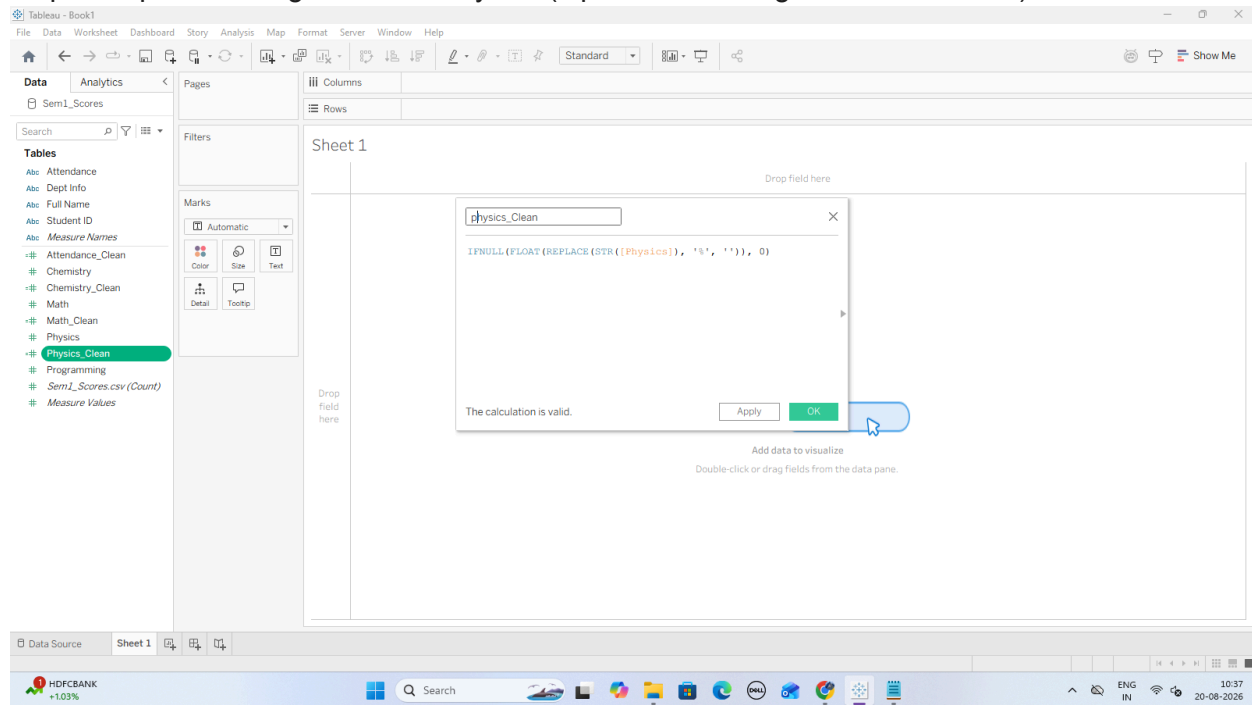
Step 3: Handle Missing (Null) Numeric Values via Imputation



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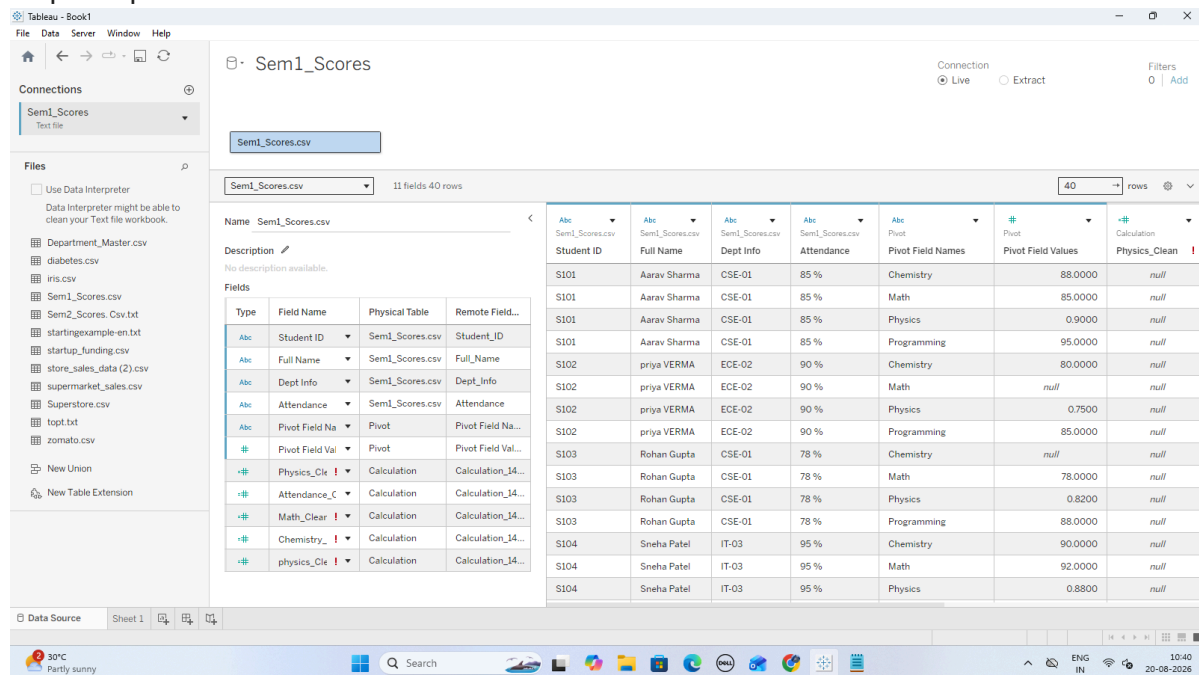
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Step 4: Impute Missing Values in Physics (Optional Handling for Student S105)



B: Convert Wide Data into Tall Format

Step 1: Open the Data Source Grid



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Step 2: Select multiple columns and choose **Pivot**; Tableau consolidates the selected columns into two fields, **Pivot Field Names** and **Pivot Field Values**, which can then be renamed as needed.

Tableau - Book1

File Data Server Window Help

Connections: Sem1_Scores (Text file)

Files: Department_Master.csv, diabetes.csv, iris.csv, Sem1_Scores.csv, Sem2_Scores_Csv.txt, startingexample-en.txt, startup_funding.csv, store_sales_data (2).csv, supermarket_sales.csv, Superstore.csv, topt.txt, zomato.csv, New Union, New Table Extension

Sem1_Scores

Sem1_Scores.csv

11 fields 40 rows

Name	Sem1_Scores.csv
Student ID	Full Name
Dept Info	Attendance
Subject	Score
Score	Physics_Clean
Attendance_C	

Student ID	Full Name	Dept Info	Attendance	Subject	Score	Physics_Clean
S101	Aarav Sharma	CSE-01	85 %	Chemistry	88.0000	null
S101	Aarav Sharma	CSE-01	85 %	Math	85.0000	null
S101	Aarav Sharma	CSE-01	85 %	Physics	0.9000	null
S101	Aarav Sharma	CSE-01	85 %	Programming	95.0000	null
S102	priya VERMA	ECE-02	90 %	Chemistry	80.0000	null
S102	priya VERMA	ECE-02	90 %	Math	null	null
S102	priya VERMA	ECE-02	90 %	Physics	0.7500	null
S102	priya VERMA	ECE-02	90 %	Programming	85.0000	null
S103	Rohan Gupta	CSE-01	78 %	Chemistry	null	null

Step 3: Clean and Format the Tall Score Field Because raw Physics contained % symbols, the new combined Score column defaults to text (Abc).

Tableau - Book1

File Data Server Window Help

Connections: Sem1_Scores (Text file)

Files: Department_Master.csv, diabetes.csv, iris.csv, Sem1_Scores.csv, Sem2_Scores_Csv.txt, startingexample-en.txt, startup_funding.csv, store_sales_data (2).csv, supermarket_sales.csv, Superstore.csv, topt.txt, zomato.csv, New Union, New Table Extension

Sem1_Scores

Sem1_Scores.csv

11 fields 40 rows

Name	Sem1_Scores.csv
Student ID	Full Name
Dept Info	Attendance
Subject	Score
Score	Physics_Clean
Attendance_C	

Student ID	Full Name	Dept Info	Attendance	Subject	Score	Physics_Clean
S101	Aarav Sharma	null	85 %	Chemistry	88.0000	null
S101	Aarav Sharma	null	85 %	Math	85.0000	null
S101	Aarav Sharma	null	85 %	Physics	0.9000	null
S101	Aarav Sharma	null	85 %	Programming	95.0000	null
S102	priya VERMA	null	90 %	Chemistry	80.0000	null
S102	priya VERMA	null	90 %	Math	null	null
S102	priya VERMA	null	90 %	Physics	0.7500	null
S102	priya VERMA	null	90 %	Programming	85.0000	null
S103	Rohan Gupta	null	78 %	Chemistry	null	null
S103	Rohan Gupta	null	78 %	Math	78.0000	null
S103	Rohan Gupta	null	78 %	Physics	0.8200	null
S103	Rohan Gupta	null	78 %	Programming	88.0000	null
S104	Sneha Patel	null	95 %	Chemistry	90.0000	null
S104	Sneha Patel	null	95 %	Math	92.0000	null

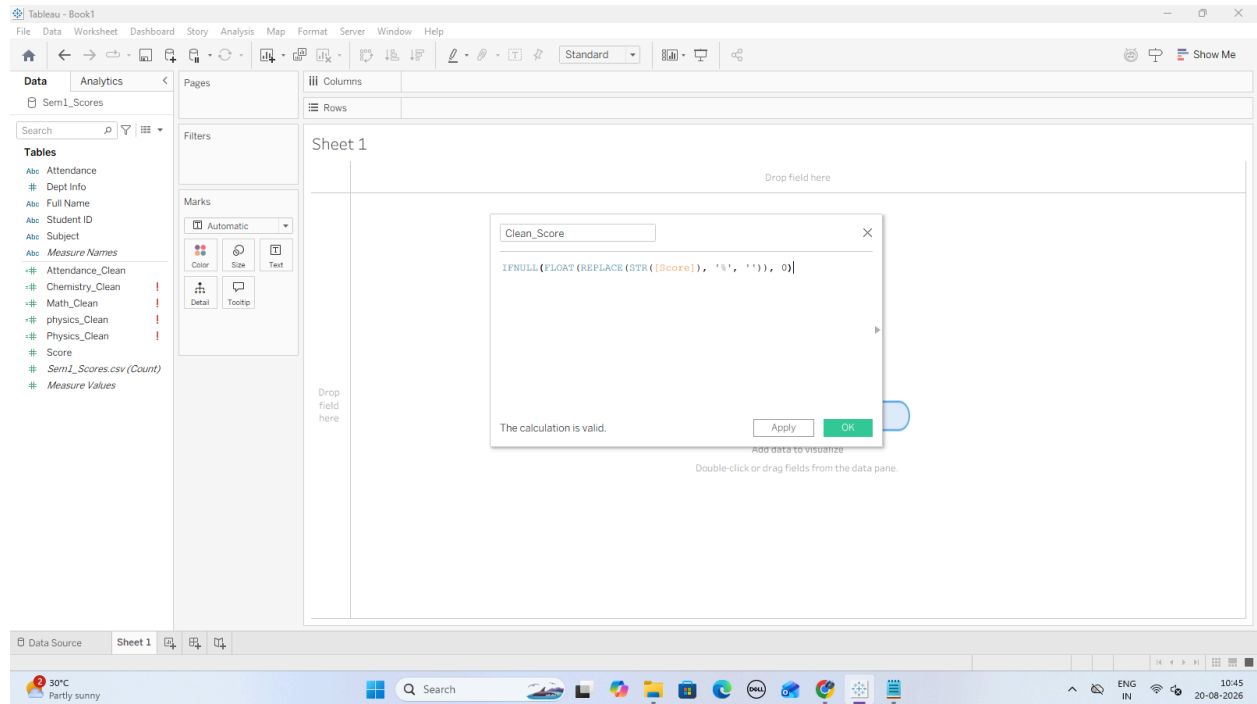
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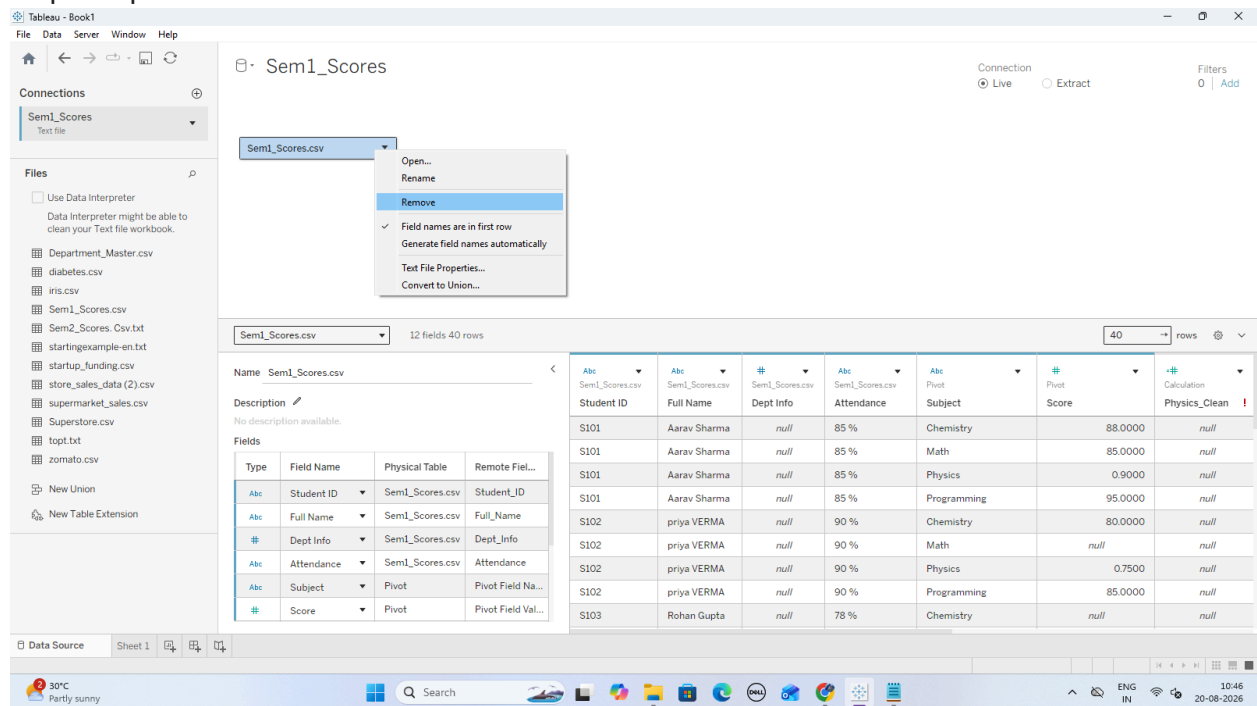
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C: Perform Union of Multiple Files Step 1: Open the Data Source Canvas



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Step 2: Create a Multi-File Union

The screenshot shows the Tableau Desktop interface with a workbook named 'Tableau - Book1'. The 'Connections' pane on the left lists several data sources, including 'Sem1_Scores' (Text file). The 'Files' pane shows a list of files, including 'Department_Master.csv', 'diabetes.csv', 'iris.csv', 'Sem1_Scores.csv', 'Sem2_Scores.csv.txt', 'startingexample-en.txt', 'startup_funding.csv', 'store_sales_data (2).csv', 'supermarket_sales.csv', 'Superstore.csv', 'topt.txt', and 'zomato.csv'. The 'Union' button is highlighted in the 'Connections' pane. The 'Fields' pane shows a list of fields, including 'Student_ID', 'Full_Name', 'Dept_Info', 'Math', 'Physics', 'Chemistry', 'Programming', and 'Attendance'. The 'Data Source' pane shows a list of data sources, including 'Sem1_Scores'.

Student_ID	Full_Name	Dept_Info	Math	Physics	Chemistry	Programming	Attendance
S101	Aarav Sharma	CSE-01	85	0.900000	88	95	85 %
S102	priya VERMA	ECE-02	null	0.750000	80	85	90 %
S103	Rohan Gupta	CSE-01	78	0.820000	null	88	78 %
S104	Sneha Patel	IT-03	92	0.880000	90	96	95 %
S105	aditya Rao	MECH-04	45	null	50	60	65 %
S106	Ananya Singh	CSE-01	88	0.910000	85	92	88 %
S107	Rahul Nair	ECE-02	null	0.700000	75	80	72 %
S108	Kavya Reddy	IT-03	95	0.940000	92	98	98 %
S109	Vikram Joshi	CSE-01	67	0.600000	58	72	80 %
S110	Pooja Iyer	ECE-02	82	0.850000	80	89	91 %

Step 3: Verify the Merged Data Grid

Step 4: Create a Clean Semester Identifier Dimension

The screenshot shows the Tableau Desktop interface with a workbook named 'Tableau - Book1'. The 'Data Source' pane shows a list of data sources, including 'Sem1_Scores'. The 'Columns' and 'Rows' panes are empty. The 'Marks' pane shows a list of fields, including 'Automatic', 'Color', 'Size', 'Text', 'Detail', and 'Tooltip'. The 'Fields' pane shows a list of fields, including 'Student_ID', 'Full_Name', 'Dept_Info', 'Math', 'Physics', 'Chemistry', 'Programming', and 'Attendance'. The 'Data Source' pane shows a list of data sources, including 'Sem1_Scores'. A dialog box titled 'Semester' is open, showing a calculated field definition:

```
IF CONTAINS([Table Name], "Sem1") THEN "Semester 1"
ELSEIF CONTAINS([Table Name], "Sem2") THEN "Semester 2"
ELSE "Other"
END
```

The dialog box also includes a 'The calculation is valid.' message and 'Apply' and 'OK' buttons. A blue arrow points to the 'OK' button.

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D: Execute Cross-Database Joins

Tableau - Book1

File Data Server Window Help

Connections

Sem1_Scores
Text file

Files

Use Data Interpreter
Data Interpreter might be able to clean your Text file workbook.

Depart...er.csv
diabetes.csv
iris.csv
Sem1_S...es.csv
Sem2_S...sv.txt
startu...e-en.txt
startu...ing.csv
store_s...(2).csv
superm...es.csv
Superstore.csv
topt.txt
zomato.csv

New Union
New Table Extension

Sem1_Scores

1 Alert

Union

Department_Master.csv

Connection
Live Extract

Filters
0 Add

Department_Master.csv 5 fields 15 rows 15 rows

Name Department_Master.csv

Description
No description available.

Fields

Type	Field Name	Physical Table	Remote Field Name
Abc	Dept Code	Department_Master.csv	Dept_Code
Abc	Department Name	Department_Master.csv	Department_Name
Abc	HOD Name	Department_Master.csv	HOD_Name
Abc	Dept Target Average	Department_Master.csv	Dept_Target_Average
#	Total Dept Capacity	Department_Master.csv	Total_Dept_Capacity

Data preview unavailable

Data Source Sheet 1 Sheet 2

30°C Partly sunny

Search

ENG IN 10:54 20-08-2026

Step 1: Stay on the Top Canvas

Step 2: Add and Link Department_Master.csv

Step 3: Add and Link Student_Activities.csv

Tableau - Book1

File Data Server Window Help

Connections

Sem1_Scores
Text file

Files

Use Data Interpreter
Data Interpreter might be able to clean your Text file workbook.

Depart...r.csv
diabetes.csv
iris.csv
Sem1_...s.csv
Sem2_...s.csv
start...en.txt
start...g.csv
store...).csv
Stude...s.csv
super...e.csv
topt.txt

New Union
New Table Extension

Sem1_Scores (2)

Department_Master.csv

Student_Activities.csv

Connection
Live Extract

Filters
0 Add

Union — Student_A...

Student_Activities.csv

How do relationships differ from joins? Learn more

Union Operator Student_Activities...

Abc Student_ID = Abc Student ID (Stu...

Add more fields

Performance Options

Remove Relationship

Abc Student_Activities.csv	Abc Student_Activities.csv	# Student_Activities.csv	# Student_Activities.csv
Student ID (Student Activ	Club Membership	Projects Completed	Certifications
S101	Coding Club	3	2
S102	Robotics Club	2	1
S103	Coding Club	2	1
S104	Debate Society	4	3
S105	Sports Club	1	0
S106	Coding Club	3	2
S107	Robotics Club	2	1

Data Source Sheet 1 Sheet 2

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Search

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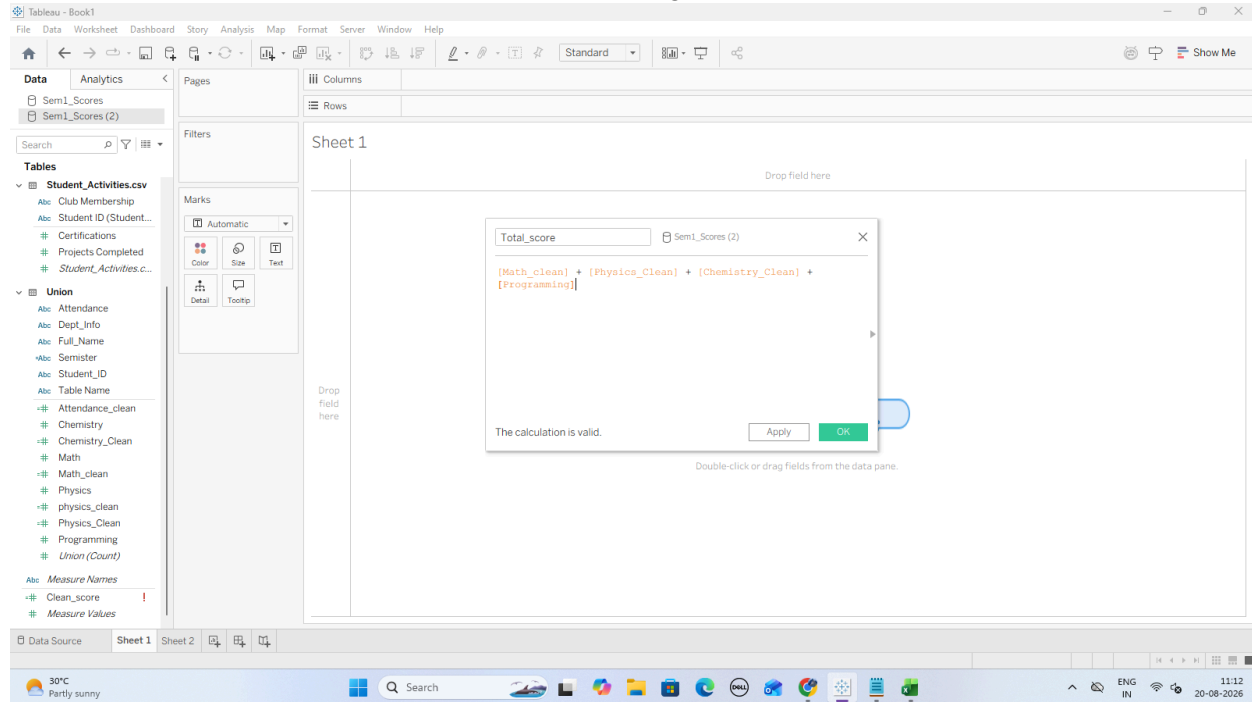
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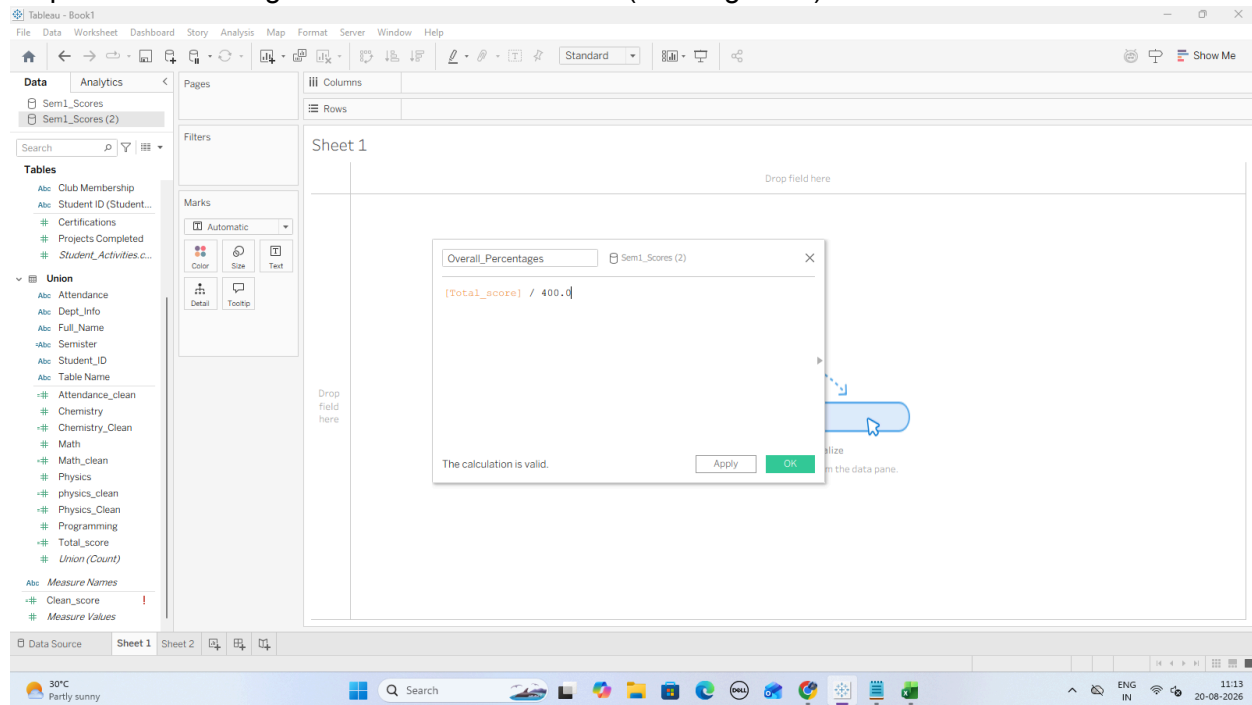
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E: Restructure Data for Better Visualization

Step 1: Calculate Total Marks and Scaled Percentage



Step 2: Create Categorical Performance Bands (Grading Tiers)



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The image displays two screenshots of the Tableau Desktop interface, showing the configuration of a calculated field for performance bands.

Top Screenshot: The 'Default Number Format [Overall_Percentages]' dialog box is open. The 'Percentage' format is selected. The 'Decimal places' are set to 1. The 'OK' button is highlighted.

Bottom Screenshot: The 'Performance_Band' calculated field dialog box is open. The formula is: `IF [Overall_Percentages] >= 0.85 THEN "Distinction"
ELSEIF [Overall_Percentages] >= 0.75 THEN "First Class"
ELSEIF [Overall_Percentages] >= 0.60 THEN "Second Class"
ELSE "Needs Academic Support"
END`. The 'OK' button is highlighted.

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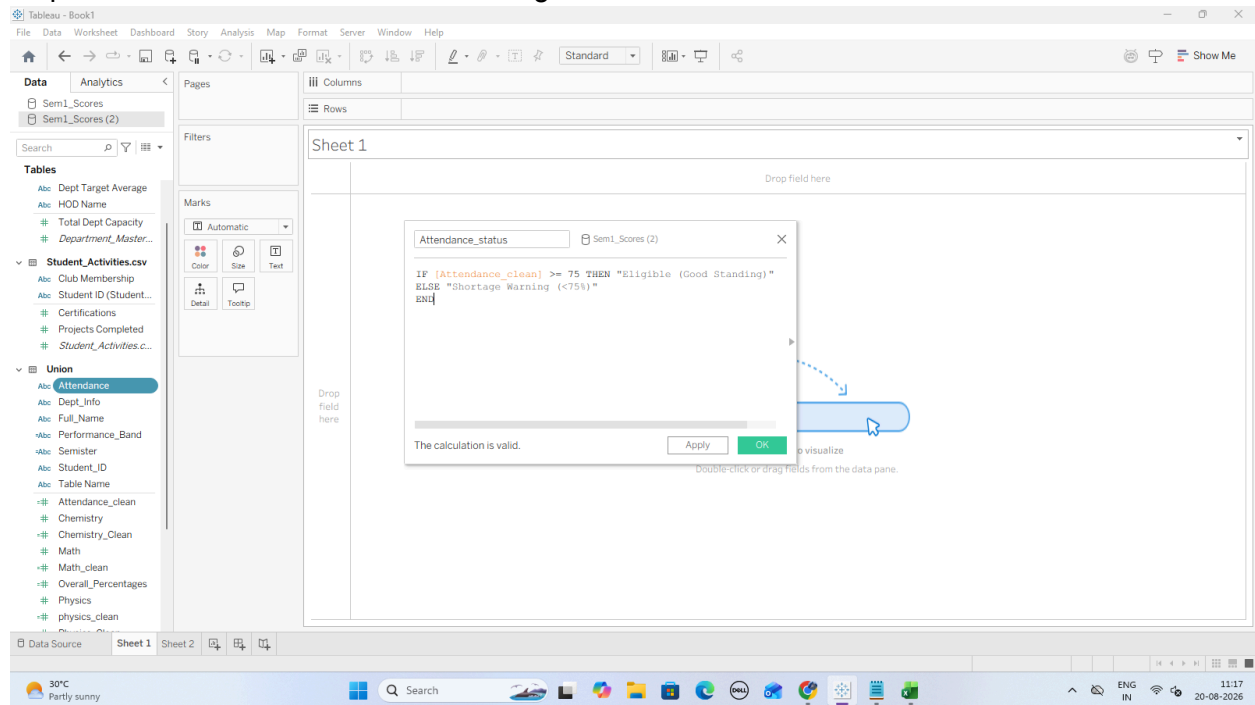
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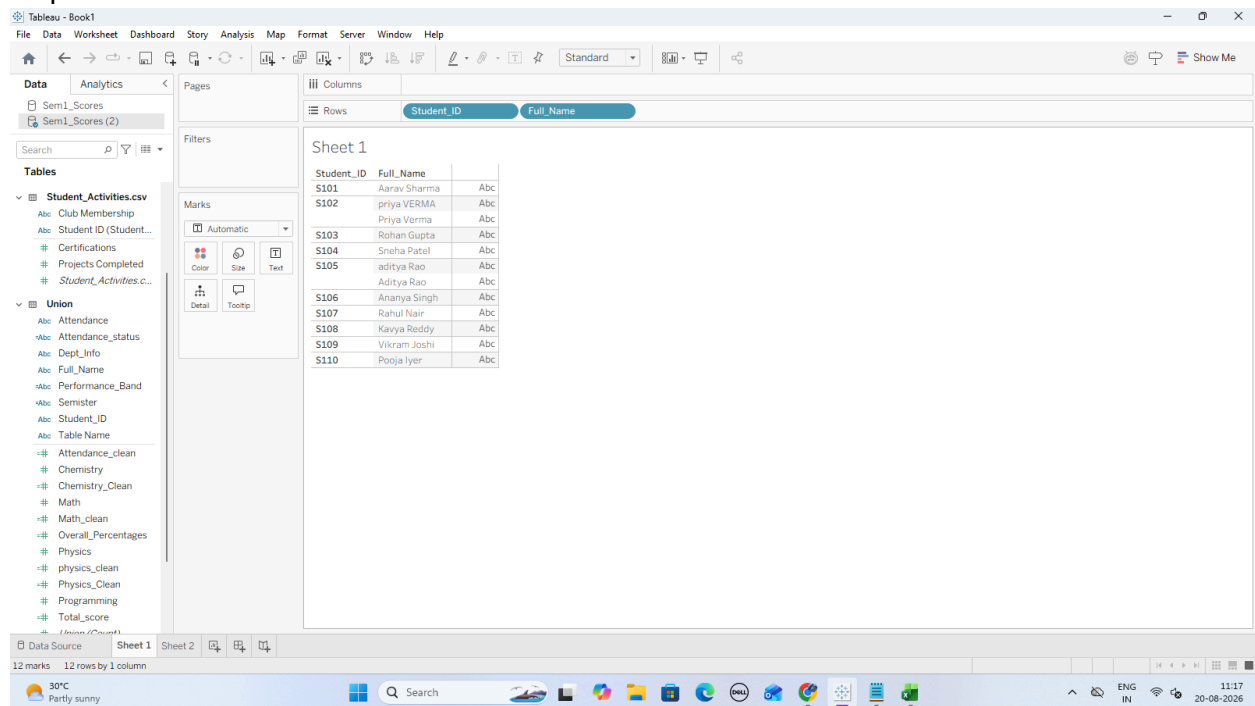
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Step 3: Create an Attendance Health Flag



Step 4: Build the Student Performance Scorecard View



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F: Handle Different Levels of Detail in Datasets

Step 1: Compute Department-Level Benchmark using {FIXED} LOD

The screenshot shows the Tableau Desktop interface. The 'Columns' shelf contains 'Student_ID' and 'Full_Name'. The 'Rows' shelf is empty. The 'Marks' shelf is set to 'Automatic'. A dialog box is open, showing the calculation 'Dept_Average_Percentage_LOD' with the formula: `{ FIXED [Dept_Info] : AVG([Overall_Percentages]) }`. The dialog box also shows 'The calculation is valid.' and 'Apply' and 'OK' buttons.

Student_ID	Full_Name
S101	Aarav Sharma
S102	priya VERMA
S103	Rohan Gupta
S104	Sneha Patel
S105	aditya Rao
S106	Ananya Singh
S107	Rahul Nair
S108	Kavya Reddy
S109	Vikram Joshi
S110	Pooja Iyer

Step 2: Compute Student Semester Average using {FIXED} LOD

The screenshot shows the Tableau Desktop interface. The 'Columns' shelf contains 'Student_ID' and 'Full_Name'. The 'Rows' shelf is empty. The 'Marks' shelf is set to 'Automatic'. A dialog box is open, showing the calculation 'Student_Overall_Career_Average' with the formula: `{ FIXED [Student_ID] : AVG([Overall_Percentages]) }`. The dialog box also shows 'The calculation is valid.' and 'Apply' and 'OK' buttons.

Student_ID	Full_Name
S101	Aarav Sharma
S102	priya VERMA
S103	Rohan Gupta
S104	Sneha Patel
S105	aditya Rao
S106	Ananya Singh
S107	Rahul Nair
S108	Kavya Reddy
S109	Vikram Joshi
S110	Pooja Iyer

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Step 3: Calculate Performance Variance (Student vs. Department Benchmark)

The screenshot shows the Tableau interface with a data source named 'Sem1_Scores (2)' loaded. The columns are 'Student_ID' and 'Full_Name'. A dialog box is open for creating a new calculation, with the formula $[\text{Overall_Percentages}] - [\text{Dept_Average_Percentage_LOD}]$ entered. The dialog also shows a preview of the data table with columns 'Student_ID', 'Full_Name', and 'Variance_From_Dept_Average'.

Student_ID	Full_Name	Variance_From_Dept_Average
S101	Aarav Sharma	Abc
S102	priya	Priya
S103	Roh	
S104	Sne	
S105	aditya	
S106	Ana	
S107	Rah	
S108	Kay	
S109	Vikr	
S110	Poo	

Step 4: Create a Benchmark Indicator Dimension

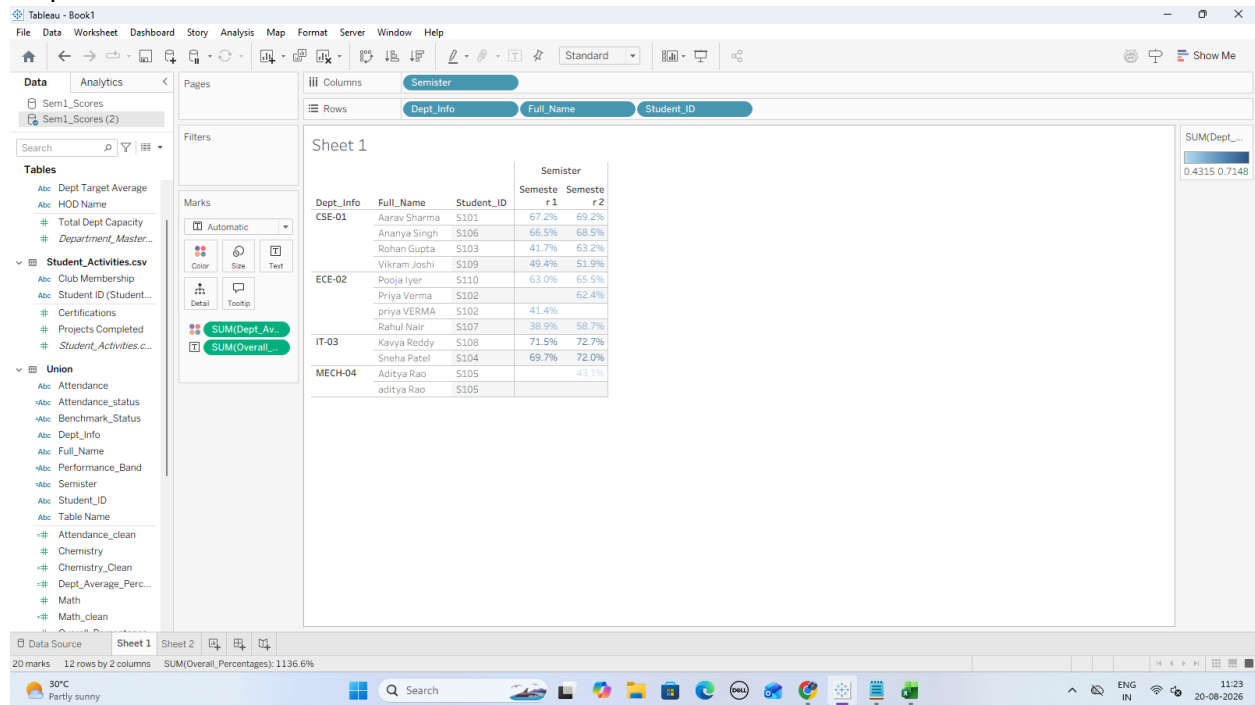
The screenshot shows the Tableau interface with the same data source. A dialog box is open for creating a new calculation, with the formula $\text{IF } [\text{Variance_From_Dept_Average}] \geq 0 \text{ THEN "Above Department Benchmark" ELSE "Below Department Benchmark" END}$ entered. The dialog also shows a preview of the data table with columns 'Student_ID', 'Full_Name', and 'Benchmark_Status'.

Student_ID	Full_Name	Benchmark_Status
S101	Aarav Sharma	Abc
S102	priya	Priya
S103	Rohan	
S104	Sneha	
S105	aditya	
S106	Anany	
S107	Rahul	
S108	Kavya	
S109	Vikrant	
S110	Pooja	

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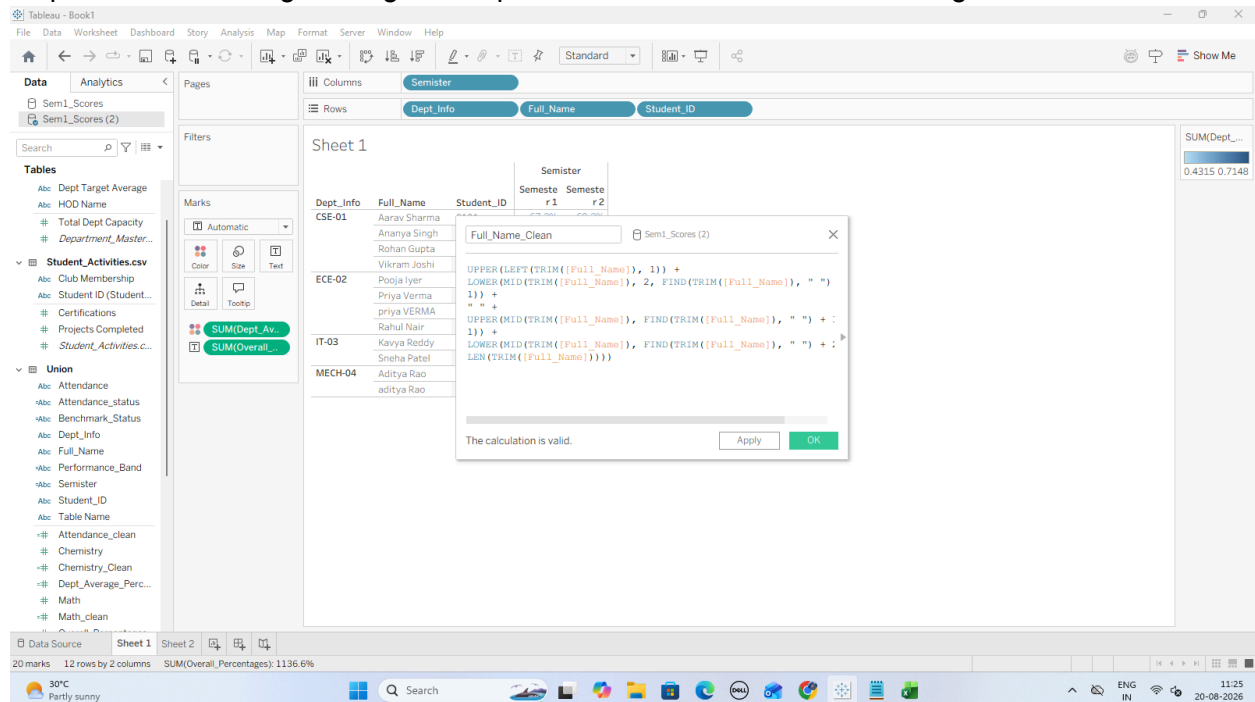
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Step 5: Build the LOD Benchmark Visualization



G: Apply Advanced Data Cleaning Techniques

Step 1: Clean Leading/Trailing Whitespace and Inconsistent Name Casing



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Step 2: Parse First Name and Last Name using SPLIT()

The screenshot shows the Tableau Desktop interface. In the Columns shelf, 'Semester' is placed. In the Rows shelf, 'Dept_Info', 'Full_Name', and 'Student_ID' are placed. A dialog box is open for the 'Full_Name' field, showing the formula `TRIM(SPLIT([Full_Name], " ", 1))`. The dialog also shows a preview of the data with columns for 'Dept_Info', 'Full_Name', and 'Student_ID'. The data is organized by department: CSE-01, ECE-02, IT-03, and MECH-04. The status bar at the bottom indicates '20 marks' and '12 rows by 2 columns'.

The screenshot shows the Tableau Desktop interface. In the Columns shelf, 'Semester' is placed. In the Rows shelf, 'Dept_Info', 'Full_Name', and 'Student_ID' are placed. A dialog box is open for the 'Full_Name' field, showing the formula `TRIM(SPLIT([Full_Name], " ", -1))`. The dialog also shows a preview of the data with columns for 'Dept_Info', 'Full_Name', and 'Student_ID'. The data is organized by department: CSE-01, ECE-02, IT-03, and MECH-04. The status bar at the bottom indicates '20 marks' and '12 rows by 2 columns'.

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Step 3: Split Composite Department Codes (Dept_Info)

The first screenshot shows the Tableau interface with the following setup:

- Columns:** Semester (Semester 1, Semester 2)
- Rows:** Dept_Info, Full_Name, Student_ID
- Calculated Field:** `SPLIT([Dept_Info], "-", 1)`
- Aggregations:** SUM(Dept_Avg), SUM(Overall_Percentage)

The second screenshot shows the final result with the department code split into two semesters and the data table populated with student records:

Dept_Info	Full_Name	Student_ID	Semester 1	Semester 2
CSE-01	Aarav Sh			
CSE-01	Ananya S			
CSE-01	Rohan Gu			
CSE-01	Vikram J			
ECE-02	Pooja Iye			
ECE-02	Priya Ver			
ECE-02	priya VEF			
IT-03	Rahul Na			
IT-03	Kavya Re			
IT-03	Sneha Pa			
MECH-04	Aditya R			
MECH-04	aditya Re			

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Step 4: Create a Clean Composite Student Label

Tableau Desktop interface showing the creation of a composite student label. The main view displays a table with columns: Dept_Info, Full_Name, Student_ID, and Semister. A calculated field 'Student_Display_Label' is being created using the formula: [Dept_Info] & " - " & [Full_Name_Clean] & " (" & [Dept_Branch] & ") " & [Semister]. An error message 'The calculation contains errors' is displayed, indicating a syntax issue with the formula.

Step 5: Verify in the Final Dashboard View

Tableau Desktop interface showing the final dashboard view. The main view displays a table with columns: Student_Display_Label, Semeste r 1, and Semeste r 2. The table shows student performance data across different departments and semesters. The calculated field 'Student_Display_Label' is verified, showing the correct composite labels for each student.

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